

Crystalline coordination framework endowed with dynamic gate-opening behaviour by being downsized to a thin film

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The fabrication of porous coordination frameworks in thin-film forms has been investigated intensively with a view to using their structural response to external stimuli and guests for potential nanotechnological applications, for example as membranes for gas separation. Here we report a coordination framework that exhibits a dynamic guest-sorption behaviour in a nanometre-sized thin-film form (16 nm thick), yet shows no guest uptake in the bulk. Highly oriented crystalline thin films of this coordination framework—which consists of interdigitated two-dimensional layers of {Fe(py)₂[Pt(CN)₄]} (py, pyridine)—were fabricated through liquid-phase layer-by-layer synthesis. The resulting thin film exhibited a clear guest uptake with a structural transformation of the gate-opening type as characterized by *in situ* X-ray diffraction. Increasing the film's thickness markedly suppressed this behaviour. We envisage that such a crystal-downsizing effect may be observed with other coordination frameworks, and may be of use to develop functional materials, for example, for switching or sensing devices.

Structural characteristics of surfaces play an important role in the functional response of living systems to external stimuli such as light, temperature, electric fields and pH. Some examples of such stimuli-responsive surfaces include the nanostructures in the super-hydrophobic lotus leaf, the optical effects of the photonic crystal structure on beetle coloration and the stimuli-activated gating of ion channels located on the cell surface^{1–5}. To understand how the functional properties or processes of chemical reactions correlate with unique surface or interface structures that differ from the internal (bulk) regions, surface chemistry that utilizes several different approaches has been advanced successfully over the past decade. Taking inspiration from biological systems, the development of biomimetic materials, surfaces based on organic polymers and mechanically interlocked molecules with stimuli-responsive properties has attracted significant attention^{6–9}.

Among the promising candidates for intelligent nanomaterials on surfaces, coordination framework materials, typically referred to as metal–organic frameworks (MOFs), and porous coordination polymers have been researched intensively^{10–15}. The variety of molecular building components—metal ions and organic-linker ligands—provides not only structural diversity (topology) and robustness, but also unique physical properties derived from their microporosity, such as gas sorption, electrical and proton conduction and catalytic properties^{16–19}. Whereas inorganic porous materials comprise metal–oxygen bonds, MOFs are held together by coordination bonds between metal ions and organic linkers that are relatively

weaker than covalent bonds and reversible, which often results in a high degree of structural flexibility. In some cases, this flexibility means that structural transformations can be triggered by external stimuli, such as the sorption of a specific guest²⁰. One example of the guest-induced switching property is the ‘gate-opening phenomenon’: from a specific vapour pressure of guest molecules (the gate-opening pressure), the sorption of those guest molecules promotes the rearrangement of the crystal structure of the host framework from a ‘closed form’ (which does not take up guests) to an ‘open form’ (which does)^{21–24}. This rearrangement is typically a sharp structural transition, and such gate-opening behaviour is often observed in relatively flexible frameworks. For rigid three-dimensional (3D) frameworks (Fig. 1a), the structural response is relatively insensitive to adsorbate loading, whereas characteristically a dynamic first-order structural transformation with hysteresis occurs at the gate-opening pressure in structurally flexible MOFs, such as 2D interdigitated frameworks and interpenetrated frameworks (Fig. 1b).

To develop an efficient use of MOFs for future nanotechnological applications, including catalysis, sensing and separation by gas-permeable membranes, research on the fabrication, characterization and application of MOF thin films has expanded rapidly in recent years^{25–29}. Among the MOF thin-film fabrication techniques proposed, three are well established bottom-up approaches for fabricating nanometre-sized MOF thin films: the layer-by-layer (LbL) method, a combination of the Langmuir–Blodgett (LB) and LbL

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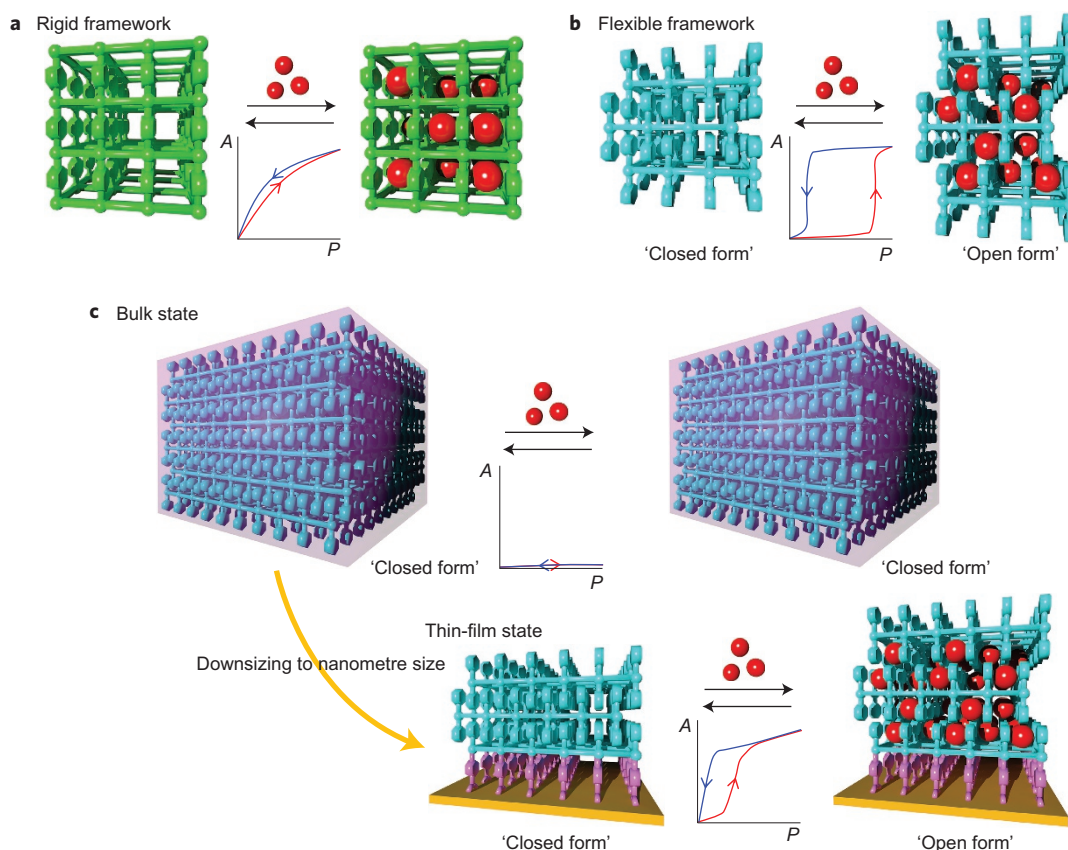


Figure 1 | Schematic representation of the various structural responses in different types of coordination frameworks. The guest molecules are denoted by red spheres, and typical examples of the sorption isotherm are also shown in the middle parts of each panel (A, guest uptake; P, gas pressure; red line, adsorption profile; blue line, desorption profile). **a**, Sorption profile typically observed for a rigid 3D framework, characterized by a gradual guest uptake with little (or no) hysteretic adsorption. **b**, Gate-opening-type guest uptake observed in a flexible 2D interdigitated framework. Abrupt adsorption occurs at the specific pressure threshold, and a large hysteresis is observed on adsorption/desorption. **c**, Dynamic structural response to guests induced by crystal downsizing from the bulk state to the nanometre-sized thin film for MOF 1. In this case, the bulk material shows no significant guest uptake, whereas in a thin film, a dynamic gate-opening response to the guest is observed.

(LB-LbL) methods, and the modular assembly method^{30–38}. The LbL method is a step-by-step approach in a liquid phase using structural components and self-assembled monolayer (SAM)-functionalized substrates, in which the substrates are alternately immersed into solutions of metal ions and organic ligands of the target MOFs^{30–35}. The LB-LbL and modular assembly methods are relatively new approaches for producing crystalline oriented MOF thin films. In the LB-LbL method, a 2D-layered MOF sheet is fabricated first on an aqueous solution via compression of a solution surface that contains metal ions and ligands, similar to an LB method. The step-by-step growth of the target MOF thin film with an intended thickness is then achieved by a repeated transfer of the 2D-layered MOF sheet onto the substrate (LbL)^{36,37}. In contrast, in the modular assembly method, 2D MOF flakes are first synthesized in a solvent by solvothermal synthesis and used as structural modules of the MOF thin film. These pre-synthesized flakes are placed on a water surface, where they assemble into a sheet that can then be transferred onto a substrate. The process can be repeated several times to form a MOF thin film on the substrate³⁸. The development of these MOF thin-film fabrication techniques allows for the control not only of film thickness but also of the crystalline orientation, which can be confirmed by X-ray diffraction (XRD) along both the parallel (in-plane) and perpendicular (out-of-plane) directions relative to the substrate surface. Although MOF thin films are expected to be more flexible than MOFs in the bulk state, the guest-induced switching properties of thin films have been investigated only rarely. To date, *in situ* XRD

monitoring of structural changes in flexible 3D MOF thin films under water vapour has been studied³⁹, and showed lattice shrinkage along one direction (the perpendicular direction with regards to the substrate surface); however, no information was obtained concerning changes in the direction parallel to the substrate surface. Dynamic structural transformations, such as gate opening from a closed form to an open form, have not yet been observed directly for MOF thin films.

Here we show the guest uptake of a gate-opening type in a highly oriented crystalline thin film of $\{\text{Fe}(\text{py})_2[\text{Pt}(\text{CN})_4]\}$ (1) (py, pyridine), a MOF that consists of interdigitated 2D layers. In the bulk state (particle size 135 nm), it exhibits no significant guest uptake. However, for a nanometre-sized crystalline thin film of the same MOF (crystal size, approximately $22 \times 22 \times 16 \text{ nm}^3$ estimated from XRD observation), an obvious response to the vapours of guests, accompanied by a dynamic structural transformation, was observed by *in situ* surface XRD analysis. There are several reports on crystal-downsizing effects in MOF materials, in which changes in the guest-adsorption behaviour (kinetics), crystal structure and magnetic property are discussed^{40–45}. To the best of our knowledge however, such a change in the dynamic guest uptake by downsizing from the bulk to a thin film has not been observed previously (Fig. 1c).

Results

Crystal structure and sorption properties in the bulk state. In this work, we focused on the Hofmann-type MOF $\{\text{Fe}(\text{py})_2[\text{Pt}(\text{CN})_4]\}$ (1) as a platform for fabricating thin films. The Hofmann-type MOF

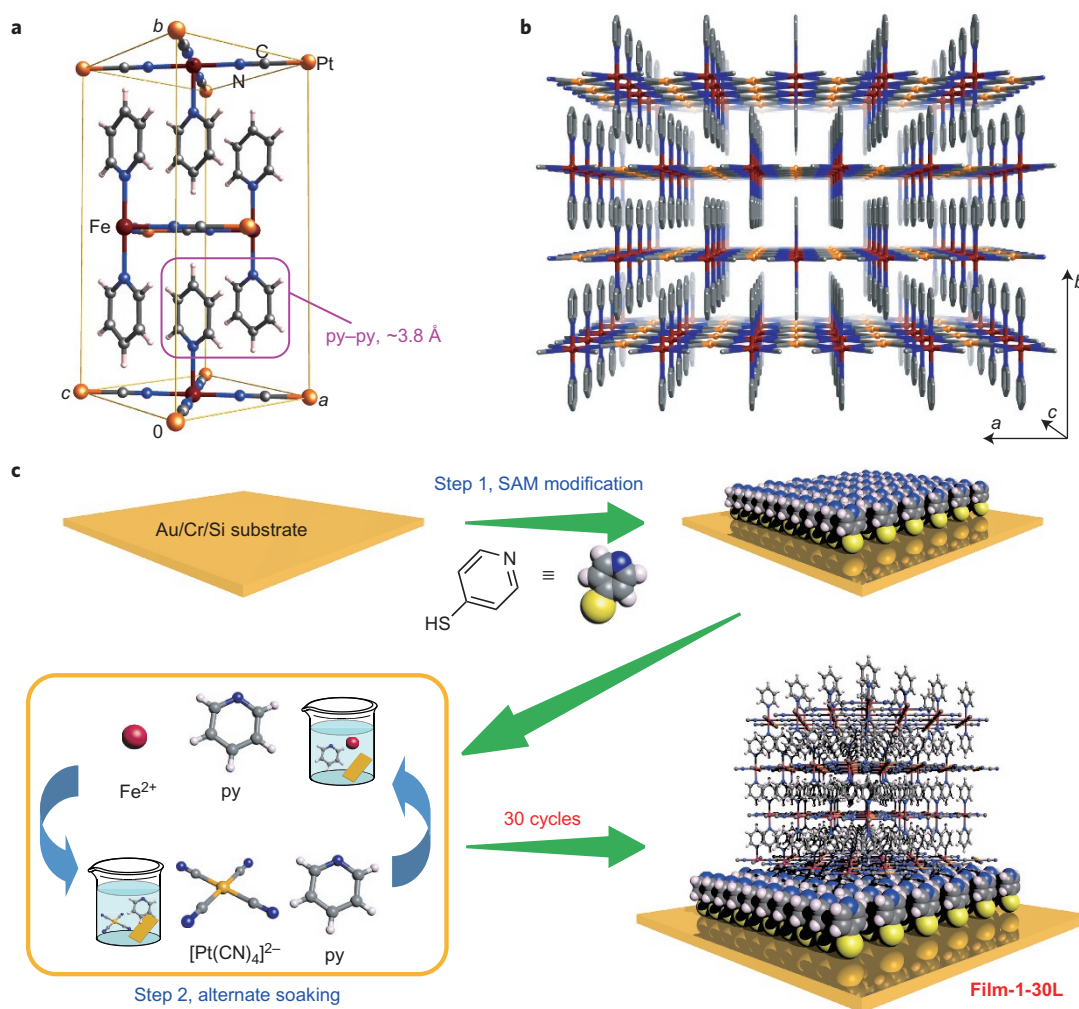


Figure 2 | Bulk-state structure of 1, and fabrication of a nanometre-sized thin film. **a,b**, Solid-state structure obtained from single-crystal X-ray crystallography (orthorhombic, *Cmcm*, $a = 7.5075(8)$, $b = 15.2605(16)$, $c = 7.4436(8)$ Å, $V = 852.80(16)$ Å³, 298 K). The unit-cell representation of **bulk-1** (**a**), in which the interplane distance between two adjacent pyridine molecules is approximately 3.8 Å, and the 3D packing structure of **bulk-1** (**b**). **c**, Schematic illustration of the thin-film fabrication of **1** based on the LbL protocol (see the text and Methods). For a clear representation of **film-1-30L**, the 4-mercaptopyridine SAM is shown as a space-filling model and the coordination framework **1** on the SAM layer is shown as a ball-and-stick model. C, grey; N, blue; H, pink; Fe, red; Pt, orange.

family is categorized as one of the earliest examples of a MOF: the frameworks consist of 2D layers that contain cyanide-bridged metal ions, which are in turn linked together by axial ligands to form overall networked structures. The first example of such a MOF, $[\text{Ni}(\text{CN})_2(\text{NH}_3)] \cdot (\text{benzene})$, was synthesized by Hofmann and Küspert in 1897⁴⁶, and its crystal structure was first revealed by XRD in 1949⁴⁷. These Hofmann-type MOFs have attracted significant attention in the investigation of a stimuli- or guest-induced cooperative spin-state change (spin-crossover phenomenon)^{48–50}. For 2D Hofmann-type MOFs, only one analogue compound of **1** with a different metal ion, $\{\text{Fe}(\text{py})_2[\text{Ni}(\text{CN})_4]\}$, has been identified structurally⁴⁸. We first prepared a single crystal of **1** (Supplementary Fig. 1) using a simple diffusion method to determine the bulk structure (**bulk-1**). Figure 2 shows the crystal structure of **bulk-1** determined by single-crystal XRD at 298 K (see also Supplementary Table 1 and Supplementary Fig. 2). In this framework, the 2D-layer structure is composed of $[\text{Pt}(\text{CN})_4]^{2-}$ units and Fe^{2+} ions, and pyridine molecules coordinate to the apical positions of each Fe^{2+} ion (Fig. 2a). The 2D layers are uniformly stacked in an interdigitated manner with π – π interaction between the pyridine molecules (Fig. 2b and Supplementary Fig. 3), with an adjacent pyridine–pyridine distance of

approximately 3.8 Å, which is similar to that in the porphyrin-based 2D MOF thin film³⁶. As, potentially, this type of framework structure is expected to exhibit a dynamical structure transformation induced by guest uptake, we investigated the sorption property of **bulk-1** using ethanol and several other adsorbates. However, no significant guest uptake was observed in the bulk state (Supplementary Figs 4–6).

Thin film fabrication and characterization. Next, we fabricated **1** as a nanometre-sized thin film (Fig. 2c) on a gold substrate using the LbL method, as described in our previous report³⁴. A Au/Cr/Si substrate covered with a SAM of pyridine-terminated thiol (4-mercaptopyridine) was soaked alternately in two different ethanol solutions that contained an excess of pyridine—one contained the Fe^{2+} species and the other $[\text{Pt}(\text{CN})_4]^{2-}$, both at room temperature—for a total of 30 cycles (Fig. 2c). In each cycle, a 2D MOF layer was introduced successively. After 30 cycles of this simple LbL protocol, a thin-film of **1** (**film-1-30L**) was fabricated successfully under ambient conditions. To monitor the thin-film growth, infrared reflection absorption spectroscopy measurements were performed after every five cycles of depositions (Supplementary Figs 7 and 8). The successive increase

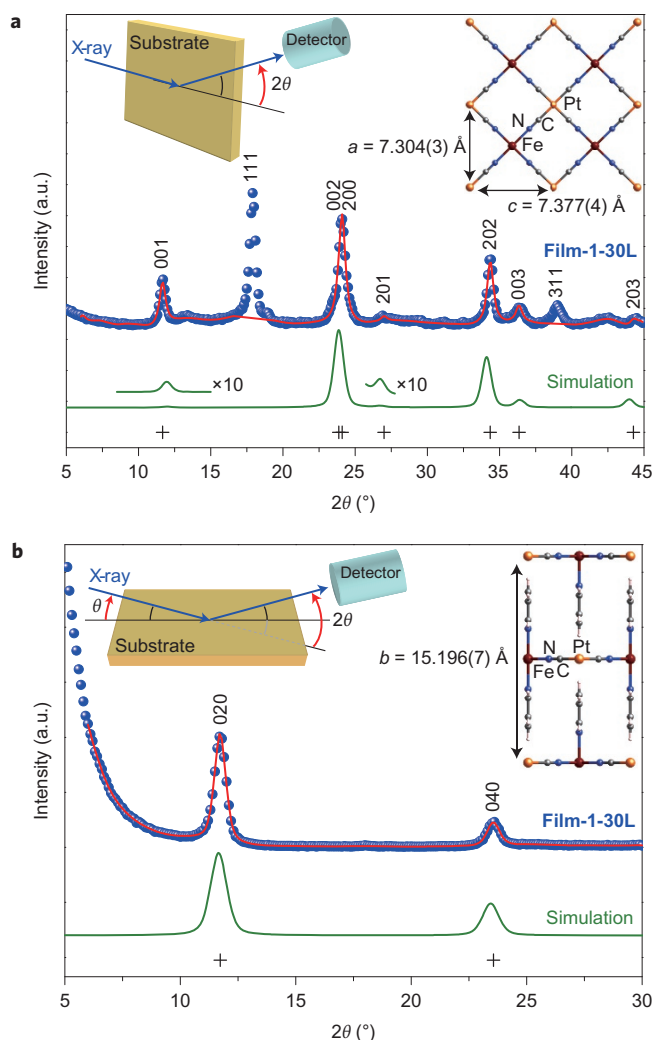


Figure 3 | Synchrotron XRD profiles of film-1-30L. **a,b**, Synchrotron XRD patterns ($\lambda = 1.550$ Å, room temperature) of **film-1-30L** in the horizontal direction (grazing incidence 2θ scan mode, in-plane (**a**)) and in the vertical direction (θ - 2θ mode, out-of-plane (**b**)). The schemes in the top left insets show the orientation of the films for each pattern (horizontal (**a**) and vertical (**b**)). The blue circles, red lines, green lines and black crosses denote the experimental data, fitting results, simulated patterns and Bragg peaks, respectively (orthorhombic, Cmmm, $a = 7.304(3)$, $b = 15.196(7)$, $c = 7.377(4)$ Å). The structures in the top right insets show the experimental geometries and periodic orderings observed in each diffraction pattern. a.u., arbitrary units.

in the intensity of the absorbance band indicates that the LbL protocol can realize a successive thin-film growth controlled by the number of deposition cycles. We also performed further characterizations of the thin film using variable-temperature XRD, Raman spectroscopy and X-ray photoelectron spectroscopy (Supplementary Figs 9–12).

Although there are many reports on the fabrication and characterization of MOF thin films, it remains challenging to create these films with a high crystallinity and well-controlled growth directions. In addition, sometimes their low electron density and thickness cause difficulties in characterization using XRD, which can provide direct structural evidence for MOF thin films. Hence, synchrotron surface XRD measurements on the obtained **film-1-30L** were performed to confirm its fabrication and investigate its crystallinity, including its orientation on the substrate surface. Figure 3 presents the synchrotron XRD patterns of **film-1-30L** both in the

parallel direction (in-plane scan, grazing-incidence 2θ scan mode (Fig. 3a)) and in the perpendicular direction (out-of-plane scan, θ - 2θ mode (Fig. 3b)) relative to the substrate surface (see the insets of Fig. 3a,b). As clearly shown in Fig. 3, the diffraction patterns observed for the two experimental geometries are quite different. Comparing the obtained diffraction patterns with the simulated XRD patterns calculated from the single-crystal X-ray structure of **bulk-1**, the diffraction peaks observed in the in-plane geometry could be indexed mainly as $h0l$, which reveals the periodicity of the 2D layer, whereas only $0k0$ peaks, which correspond to the interlayer spacing mediated by the pyridine ligand, were observed in the out-of-plane geometry. However, two peaks assignable to the 111 and 311 diffraction peaks were also observed (Fig. 3a), which would not be observed in a perfect crystalline oriented thin film. These peaks did not originate from the bare substrate components (Cr or Au). To explain the observation of these diffraction peaks, we also performed rocking-curve (θ scan mode) and azimuthal-angle-dependence (φ scan mode) measurements at the 020 Bragg peak (Supplementary Fig. 13). In the rocking-curve measurement, a very weak single peak at θ of approximately 5.9° with a full-width at half-maximum of 4.5° was observed, which suggests that each 2D layer is stacked with an average tilting angle of $\sim 4.5^\circ$. Therefore, the observed 111 and 311 peaks originate from the canted nature of **film-1-30L** on the substrate. The result of the azimuthal-angle-dependence measurement clearly indicates the uniformity of the obtained thin film. From the 020 (out-of-plane XRD) and 001 (in-plane XRD) peaks, using Scherrer's equation the average thickness and crystal domain (island) size were calculated to be approximately 16 and 22 nm, respectively. From the atomic force microscopy (AFM) image results, several island structures of this thin film (thickness, 20–40 nm; side, <30 nm and 70–100 nm) were observed (Supplementary Fig. 14).

Specific sorption properties that arise from a thin-film state. As demonstrated above, the synchrotron XRD measurements provide clear evidence that the obtained **film-1-30L** is crystalline and exhibits a preferentially oriented nature on the substrate surface. The 2D interdigitated structure is expected to enable a guest-responsive structural change in the vertical direction, although **bulk-1** shows no significant guest uptake. To investigate the characteristic response to the guest of **film-1-30L**, we then performed *in situ* XRD experiments under ethanol vapour at room temperature. *In situ* in-plane and out-of-plane XRD patterns under different ethanol relative pressure (P/P_0) conditions are presented in Fig. 4. No significant change in the in-plane diffraction pattern was observed, which indicates that the 2D layer remains unchanged before and after the introduction of ethanol (Fig. 4a and Supplementary Fig. 15). However, in Fig. 4b for the out-of-plane geometry, both the 020 and 040 peaks are clearly shifted to the lower 2θ region under the presence of ethanol vapour ($2\theta = 11.51^\circ$ for 020, where $b = 15.44$ Å at $P/P_0 = 0.93$), and the peaks are shifted back to the original position when the ethanol vapour is removed ($2\theta = 11.69^\circ$ for 020, where $b = 15.19$ Å at $P/P_0 = 0$). These changes in the diffraction profiles are completely different to those of **bulk-1** (Supplementary Fig. 4 inset) and indicate that **film-1-30L** transformed abruptly from the closed form to the open form, accompanied by the anisotropic expansion of the interlayer spacing induced by the ethanol vapour. We performed further *in situ* XRD measurements using other solvent vapours. Similar reversible peak shifts were also observed using relatively small molecules, such as water, methanol and acetonitrile, which indicates that such behaviour depends on the molecular size of the adsorbates (Supplementary Figs 16 and 17). Measurements with a quartz-crystal microbalance also confirmed guest uptake in **film-1-30L** as successive weight changes (Supplementary Figs 18–20). For

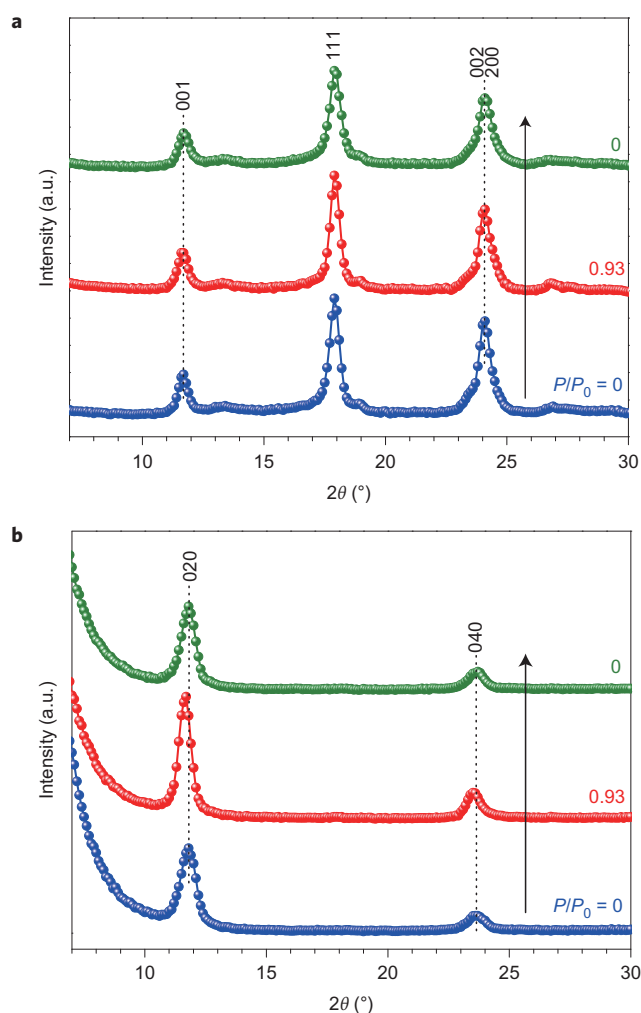


Figure 4 | Anisotropic structural change of film-1-30L induced by guest introduction. **a,b**, In-plane (**a**) and out-of-plane (**b**) XRD profiles of film-1-30L with varying relative pressure (P/P_0) of ethanol vapour ($\lambda = 1.550 \text{ \AA}$, room temperature), where P_0 is the saturated vapour pressure of ethanol at room temperature (7.833 kPa). The diffraction patterns were collected continuously at three pressure points in the following order (vertical arrows): before the introduction ($P/P_0 = 0$, blue), at $P/P_0 = 0.93$ (red) and after the removal ($P/P_0 = 0$, green) of ethanol vapour.

comparison, we also investigated changes in the sorption behaviour by crystal downsizing of **bulk-1**. Simple mechanical grinding of the as-synthesized powder of **bulk-1** (average particle size 135 nm) using a mortar afforded small nanocrystals of **1** (average particle size 8.9 nm (Supplementary Fig. 21)). When the crystal size decreased, vapour-sorption behaviour was observed clearly, which supports the crystal-downsizing effect of **1** (Supplementary Fig. 22).

Confirmation of gate-opening behaviour and downsizing effect.

To clarify the details of this unique phenomenon observed in film-1-30L, we performed additional *in situ* out-of-plane XRD measurements by successively varying the relative pressure of ethanol (Fig. 5a,b). Both the 020 and 040 peaks gradually shifted with increasing relative pressure, and vice versa. The change in interlayer spacing (which corresponds to the lattice parameter b) calculated from the 020 and 040 peak positions enables the illustration of the clear closed-to-open dynamic structural transformation in film-1-30L (Fig. 5c). As observed in Fig. 5c, for the adsorption process, film-1-30L drastically expands in the vertical direction (parallel to the b axis) to the substrate in the

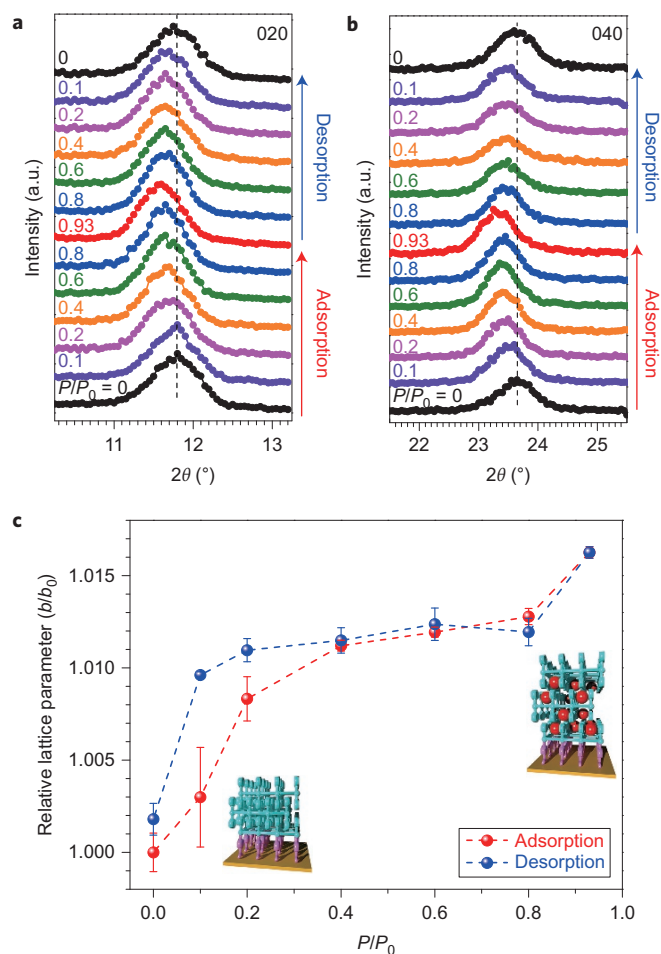


Figure 5 | Dynamic closed-to-open structural transformation of film-1-30L under ethanol vapour. **a,b**, The *in situ* out-of-plane XRD profiles of film-1-30L under various ethanol vapour pressures in the 020 (**a**) and 040 (**b**) regions at room temperature. **c**, Relative lattice constant (b/b_0) calculated from the 020 and 040 peak positions plotted against the relative pressure (P/P_0) of ethanol vapour, where b_0 is the lattice constant of film-1-30L before guest uptake ($P/P_0 = 0$, black circles in **a** and **b**). The red and blue circles represent the adsorption and desorption processes, respectively. Error bars denote $\pm$ s.d. using two b/b_0 values calculated from the 020 and 040 diffraction peaks. The inset figures show schematic illustrations of the response of film-1-30L to the presence of ethanol guests, obtained from *in situ* X-ray observations, with the MOF, SAM and guest molecules shown in light blue, pink and red, respectively.

lower pressure region below $P/P_0 = 0.4$ and then gradually expands and reaches the largest interlayer spacing at $P/P_0 = 0.93$. For the desorption process, there is a small hysteresis in the lower pressure region ($P/P_0 \leq 0.4$), which indicates that this dynamical lattice expansion in film-1-30L is a first-order structural transformation. A similar, but small, reversible interlayer-spacing change was also detected when using acetonitrile as an adsorbate, which indicates that the dynamical structural transformation of film-1-30L depends on the guest species (Supplementary Fig. 23). The observed lattice expansions by guest introduction along the b axis were small (approximately 1.6% under ethanol vapour, and approximately 0.5% under acetonitrile vapour). Focusing on the crystal structure of the as-synthesized **bulk-1**, a capsule-shaped pore with the dimensions $2.9 \times 3.0 \times 5.2 \text{ \AA}^3$ is formed between the 2D layers (Supplementary Fig. 24), a size suitable for the accommodation of small guest molecules (Supplementary Figs 25 and 26). Therefore, it is suggested that **1** can be inherently porous,

and a small lattice expansion, such as observed in this study (less than 2%), is large enough for guest adsorption.

The results obtained in this study show a switch in the guest-uptake property, accompanied by an anisotropic structural transformation from the closed form to the open one in a nanometre-sized thin film, which implies the existence of the crystal-downsizing effect in this system. To understand precisely this phenomenon, we further investigated the dependency of the thin-film size on the gate-opening behaviour. In this system, it is difficult to control the domain size of the thin film along the in-plane direction parallel to the substrate; however, the domain size along the out-of-plane direction (that is, the thickness) can be controlled using a simple LbL protocol. Three thin films of different thicknesses (**film-1-60L**, **film-1-120L** and **film-1-150L**) were fabricated successfully by varying the number of LbL cycles. The obtained thin films also exhibited crystallinity and a preferentially oriented nature, in a similar way to **film-1-30L** (Supplementary Figs 27–31). Although the domain sizes of each thin film along the in-plane direction parallel to the substrate were almost constant (19 nm for **film-1-30L**, 22 nm for **film-1-60L**, 22 nm for **film-1-120L** and 23 nm for **film-1-150L**), the average thickness of the thin film increased successively with the number of LbL cycles (16 nm for **film-1-30L**, 30 nm for **film-1-60L**, 45 nm for **film-1-120L** and 49 nm for **film-1-150L** (Supplementary Figs 32 and 33). From AFM observations, the island size along the parallel direction relative to the substrate seems to become larger and the interfaces among adjacent domains seem to become more unclear with increasing numbers of LbL cycles (thickness, 50–75 nm; side, >120 nm for **film-1-60L** (Supplementary Fig. 30)). Structural transformation of the gate-opening type observed in **film-1-30L** was suppressed by the increase of the thickness of thin film, which clearly demonstrates the existence of the effect of crystal downsizing on **1** (Supplementary Fig. 34).

Discussion

In this work, we discovered the gate-opening switching behaviour of 2D interdigitated MOF **1** realized in a nanometre-sized thin film. In addition, this dynamic structural response to the guests is not observed in the bulk-crystal state, and when the crystal size decreases down to the nanometre size, the thin-film state exhibits a selective guest uptake accompanied by anisotropic expansion along the vertical direction to the substrate surface, which is completely different from that of the bulk state. Moreover, the results of the LbL cycle dependency indicate that this type of gate-opening behaviour in the thin film arises from downsizing the film thickness. The origin of such a change in response to the guests (and the transition in an energetically stabilized guest-adsorbed phase) is not yet clear; however, in the thin-film state the diffusion length for guest molecules is decreased compared with that of the bulk crystal. The change in island morphology seen by AFM for the thicker thin films might also hinder the diffusion of guests. In the bulk state, we suggest that the guest penetration within the framework is prevented (or limited to the surface area) by the strong π – π stacking between pyridine ligands (see Fig. 2). However, the surface area is significantly increased on decreasing the crystal size down to the nanometre scale, where the ligand π – π interaction is not as effective as in the bulk because of the reduced number of ligands at the outermost surface. This situation would contribute to the lowering of the potential barrier height for a gate-opening-type guest uptake, which results in a specific response to the guests and adsorption kinetics in thin films that differs from those of bulk crystals. These findings provide insight towards the understanding of the mechanism that dominates the switchable sorption properties or host–guest interactions of MOF thin films, which may help their development for future applications, such as chemoswitching or sensing applications, gas separation, catalytic devices and integrated systems of functionality (for example, an innovative fuel cell).

Methods

Preparation of the {Fe(py)₂[Pt(CN)₄]} single crystal (**bulk-1**).

Fe(NH₄)₂(SO₄)₂·6H₂O (196 mg, 0.5 mmol), K₂[Pt(CN)₄] (189 mg, 0.5 mmol) and citric acid (451 mg, 2.5 mmol) were mixed in 60 ml of water. Then, 1,3-diaminopropane (1.0 ml, 12 mmol) was added to the solution to yield a yellow transparent solution of Fe(NH₃)₂[Pt(CN)₄]. An excess amount of pyridine (5.0 ml) was slowly diffused into H₂O that contained this yellow transparent solution (10 ml) using an H-shaped glass tube at room temperature. Block-shaped single crystals of **1** were obtained after three weeks (Supplementary Fig. 1). Elemental analysis (% calculated, % found for C₁₄H₁₀N₆FePt: C (32.77, 32.54), H (1.96, 1.91), N (16.38, 16.32)).

Single-crystal X-ray crystallography. X-ray crystal structure analysis of **bulk-1** was performed using a Bruker SMART APEX II CCD detector with graphite-monochromated Mo K α radiation (λ = 0.71073 Å) at 298 K. The details are provided in the Supplementary Information.

Preparation of the {Fe(py)₂[Pt(CN)₄]} thin film (**film-1**).

A Au/Cr/Si substrate (20 × 15 × 0.5 mm) was soaked in an ethanol solution of 4-mercaptopyridine overnight. The substrate was rinsed with ethanol and dried by N₂ flow, then soaked alternately in the ethanol solutions of the building units Fe(BF₄)₂·6H₂O (10 mM) and [(C₄H₉)₄N]₂[Pt(CN)₄] (10 mM) at room temperature for the total of the intended cycles (30–150 cycles). Both solutions contained an excess amount of pyridine (100 mM). Every soaking process was followed by an ethanol-wash treatment.

Synchrotron XRD measurements. Synchrotron XRD experiments were performed at room temperature with a multi-axis diffractometer with a scintillation detector installed on the BL13XU beamline of SPring-8, Japan. Helium gas was always supplied through the cell during the measurement. The relative vapour pressure for *in situ* XRD and grazing-incidence XRD experiments was controlled by tuning the ratio of vapour/helium using a BEL-Flow apparatus (MicrotracBEL Corp.).

Data availability. X-ray crystallographic data for **bulk-1** have been deposited in the Cambridge Crystallographic Data Centre (CCDC) under deposition number CCDC 1405911, and can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif.

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Author contributions

K.O. and H.K. conceived the work and designed this study. S.S., K.O. and C.S. performed the experiments. O.S., C.S., A.F. and M.T. contributed to the synchrotron XRD measurements. S.S., K.O. and H.K. co-wrote the manuscript. All the authors discussed and commented on the paper.

Additional information

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Competing financial interests

The authors declare no competing financial interests.